

RO Number: 3809

Inspection Number: 3180956

Year: 2020
Make: Mazda
Model: CX-5 Grand Touring AWD
VIN: JM3KFBDM2L0722940

Insurance: GEICO CASUALTY COMPANY

Estimator: Dykeman, Andrew

IMPORTANT! Please validate vehicle is equipped with listed ADAS system components by performing a visual inspection!

Scanning Requirements

PRE AND POST SCANS REQUIRED

[https://oem1stop.com/sites/default/files/Mazda_Pre-and-Post-Repair-Scanning_Position\(1-18\)%20.pdf](https://oem1stop.com/sites/default/files/Mazda_Pre-and-Post-Repair-Scanning_Position(1-18)%20.pdf)

Mazda North American Operations recommends that all vehicles being repaired for collision damage be scanned before and after the repair. Diagnostic Trouble Codes (DTCs) will be stored if any of the sensors, cameras, or radars were damaged in the collision.

Identified ADAS Related Services to be Performed

Based on the estimate, the following calibrations/initializations have been identified:

- ADAS L3: Static Calibration - A (2)
- Seat Belt Inspection (1)
- Seat Related Calibration (1)
- Steering Angle Sensor (1)

Recommendations

OPUS recommends a pre-scan on this vehicle. To determine what type of scan should be performed, refer to service information and certification requirements.

OPUS recommends utilizing OEM software to perform a post-scan on this vehicle.

Calibrations

<u>ADAS System/Components</u>	<u>Location</u>	<u>Calibration</u>
Adaptive Cruise Control		
Front Radar Sensor	Front Bumper	ADAS L3: Static Calibration - A

Calibrate the Front Radar sensor if any of the following is removed, replaced, or struck in a collision:

- radar unit
- radar bracket
- radiator support (replace radar unit if struck)

NOTE: Calibration required if bumper cover or grille are struck as it may deviate the radar sensor position.

Targets are required for calibration and wheel alignment if suspected out of spec.

OEM Reference:

[https://opusadas.z13.web.core.windows.net/links?adasInspection=3180956&adasChecklistMatrixVehicle=2665&type=1&model=CX-5 Grand Touring AWD](https://opusadas.z13.web.core.windows.net/links?adasInspection=3180956&adasChecklistMatrixVehicle=2665&type=1&model=CX-5%20Grand%20Touring%20AWD)

Blind Spot Monitoring Systems

Spot Sensors - Rear

Rear Bumper

ADAS L3: Static Calibration - A

Calibrate the Rear Side Radar System (rear blind spot sensors) when bumper is replaced or if any of the following are removed, installed, or replaced:

- Side radar unit (replace radar unit if struck and do not repair bumper over radar)
- Side radar bracket

Note: If paint coat thickness has changed, the radar emissions angle will deviate.

ADAS targets required for system calibration.

OEM Reference:

[https://opusadas.z13.web.core.windows.net/links?](https://opusadas.z13.web.core.windows.net/links?adasInspection=3180956&adasChecklistMatrixVehicle=1053&type=1&model=CX-5%20Grand%20Touring%20AWD)

[adasInspection=3180956&adasChecklistMatrixVehicle=1053&type=1&model=CX-5 Grand Touring AWD](https://opusadas.z13.web.core.windows.net/links?adasInspection=3180956&adasChecklistMatrixVehicle=1053&type=1&model=CX-5%20Grand%20Touring%20AWD)

Occupant Detection

Passenger Seat Weight Sensor

Passenger Seat

Seat Related Calibration

Important! Check for DTC and requirements before performing calibration!

The occupant classification sensor must be calibrated whenever passenger seat components or the sensor itself is replaced or disassembled.

For cases involving **passenger seat removal, reinstallation, or vehicle collision**, a **DTC inspection is first required**. During this **DTC inspection** procedure, the **system will calibrate** the sensor during self-tests. Verify seat position and environmental requirements prior to performing these self-tests.

OEM Reference:

[https://opusadas.z13.web.core.windows.net/links?](https://opusadas.z13.web.core.windows.net/links?adasInspection=3180956&adasChecklistMatrixVehicle=1693&type=1&model=CX-5%20Grand%20Touring%20AWD)

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Seat Belt Inspection

Seat Belt

Interior

Seat Belt Inspection

Seatbelts must be inspected after a collision to ensure passenger safety. While ADASMap and Alldata serve as valuable resources to assist in the repair process, it is ultimately the responsibility of the repair professional to follow proper procedures and safety guidelines. For the most accurate and up-to-date information, always consult the vehicle manufacturer's service manual when available.

OEM Reference:

[https://opusadas.z13.web.core.windows.net/links?](https://opusadas.z13.web.core.windows.net/links?adasInspection=3180956&adasChecklistMatrixVehicle=4423&type=1&model=CX-5%20Grand%20Touring%20AWD)

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Steering Angle Sensor

Steering Angle Sensor

Column

Steering Angle Sensor

Mazda has many iterations of steering angle sensors throughout the model years. The newer model years are varied between electronic power steering systems that incorporate a steering angle and torque sensor as integral parts of the EPS motor assembly. Mazda makes no clear statement on whether an alignment of the sensor must be performed on all models with EPS, but suggests a complete vehicle scan is performed to check for related DTC's for steering angle.

OEM Reference:

[https://opusadas.z13.web.core.windows.net/links?](https://opusadas.z13.web.core.windows.net/links?adasInspection=3180956&adasChecklistMatrixVehicle=4423&type=1&model=CX-5%20Grand%20Touring%20AWD)

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Identified ADAS Systems

<u>System</u>	<u>Component</u>	<u>Location</u>
Adaptive Cruise Control	Front Radar Sensor	Front Bumper
Blind Spot Monitoring Systems	Blind Spot Sensors - Rear	Rear Bumper
Lane Departure / Keep Systems	Forward Sensing Camera (FSC)	Windshield
Occupant Detection	Passenger Seat Weight Sensor	Passenger Seat
Seat Belt Inspection	Seat Belt	Interior
Steering Angle Sensor	Steering Angle Sensor	Column

The Fine Print:

The Fine Print (Legaleze)

The ADAS MAP™ report is for informational purposes only and should not be considered as a replacement for Original Equipment Manufacturers (OEM) diagnostic, repair or calibration recommendations and procedures. The ADAS MAP™ platform utilizes data from multiple sources such as VIN, ADAS and safety system build data. While this data is considered accurate, Auto Techcelerators, LLC. makes no guarantees about its accuracy or completeness because it is being generated and delivered by third-parties.